When Fusion Breaks

Graceful Degradation in Multimodal Emotion Recognition

A systematic study of how multimodal fusion architectures fail when their inputs degrade — and an open, reproducible framework for measuring it.
Dataset: CMU-MOSEI (22,856 annotated utterances) | Code: github.com/inayatarshad/Multimodal-Emotion-Recognition

Executive Summary

Multimodal emotion recognition systems combine language, voice and facial signals, and report strong accuracy on benchmarks. Those benchmarks assume every input stream is present and clean. Deployed systems do not get that: cameras drop frames, microphones clip, and automatic speech recognition delivers transcripts with substantial word error rates. A model's benchmark score therefore says very little about whether it will hold up in the field.

This project builds the missing measurement. It implements six fusion architectures spanning the range from loosely coupled to tightly coupled, subjects them to fourteen controlled corruption operators at graded severity, and quantifies degradation with metrics designed for the purpose. The central question is whether the architectures that win on clean benchmarks are the ones that fail hardest under realistic conditions — a question with direct consequences for anyone deploying these systems outside a laboratory.

> **Contribution.** The novel element is not another fusion architecture. It is a rigorous, reusable protocol for measuring robustness: corruption operators with a normalised severity scale and a bitwise-identity guarantee at zero, a chance-corrected retention metric, and an evaluation design in which every architecture is scored on bitwise-identical corrupted inputs so that differences between them are attributable to the model rather than to sampling noise.

1. Problem Statement

Benchmark evaluation of multimodal models measures performance under conditions that deployment rarely provides. Three failure modes dominate in practice:

- **Transcription error.** Speech recognition on accented, code-switched or low-resource speech produces transcripts with word error rates that can exceed 30 percent. The text stream arrives, but wrong.
- **Signal degradation.** Compressed or bandwidth-limited video loses frames; inexpensive microphones clip and add noise; faces leave the frame entirely.
- **Stream misalignment.** Audio and video are frequently buffered independently, so the temporal correspondence that fusion models rely on is not guaranteed.

Existing work on missing modalities studies complete absence of a stream, usually for a single architecture. What is absent from the literature is a systematic, graded comparison across architectures under a common protocol. Without one, a practitioner choosing a model has benchmark accuracy and nothing else to go on.

2. Research Question and Hypothesis

> **Hypothesis (H1).** More sophisticated fusion architectures are more brittle. Models that learn rich cross-modal dependencies should degrade faster under corruption than models that combine modalities loosely, because they have learned to depend on interactions that no longer exist once a stream is damaged.

The hypothesis is stated before the experiments and is falsifiable in a useful direction: if it is wrong — if tightly coupled fusion turns out to be more robust — that is a more interesting finding than a mild confirmation, and it is reported as such. Three secondary questions follow:

- **Q2 — Reliance asymmetry.** These models are widely suspected of being text-dominated. How much performance survives when text alone is removed, versus audio and vision together?
- **Q3 — Mitigation.** Does randomly dropping modalities during training buy test-time robustness, and what does it cost in clean accuracy?
- **Q4 — Graded versus binary.** Does partially degraded input behave like partial absence, or differently?

3. System Design

Six architectures are implemented behind a single interface, spanning the coupling axis that H1 treats as the independent variable. Encoder family, hidden width, optimiser and training schedule are held constant across all of them, so any measured difference is attributable to the fusion mechanism rather than to incidental capacity.

Architecture	Fusion mechanism	Coupling
Unimodal baselines	None; a single stream is used in isolation	None
Late fusion	Independent encoders; decisions combined by learned weights	Loose
Early fusion	Frame-level concatenation into one shared encoder	Moderate
LMF	Low-rank factorised outer product across modalities	Multiplicative
TFN	Full outer product; all uni-, bi- and tri-modal interaction terms	Multiplicative
MulT	Six directional cross-modal attention transformers	Tight

Table 1. Architectures on the fusion-coupling axis.

4. Degradation Protocol

Fourteen corruption operators model failures that actually occur in deployment, rather than generic noise. Each accepts a severity in the unit interval and maps it onto its own physical parameter, so a single sweep configuration drives every family and results remain comparable across them.

Stream	Operator	Severity maps to	Deployment failure modelled
Text	**ASR error channel**	word error rate, 0-40%	substitution, deletion and insertion at realistic proportions
Text	**Token dropout**	% tokens lost	unrecognised words
Text	**Word shuffle**	permutation window	syntax lost, vocabulary retained
Audio	**Additive noise**	SNR, 20-0 dB	ambient and channel noise
Audio	**Frame dropout**	% frames lost	packet loss
Audio	**Burst dropout**	% contiguous loss	a stream cutting out
Audio	**Clipping**	threshold vs RMS	microphone overload
Visual	**Occlusion**	% frames occluded	face leaves the frame
Visual	**Temporal blur**	Gaussian sigma	low frame rate, motion blur
Visual	**Frame dropout**	% frames lost	tracker loses lock
All	**Temporal shift**	frames of offset	independently buffered streams
All	**Removal (3 variants)**	fraction removed	zero-fill, mean-fill, learned mask

Table 2. Corruption operators. The full grid is 32 evaluation axes covering graded severity, temporal misalignment and all seven non-empty subsets of removed modalities.

Two design guarantees

Severity zero is a bitwise identity. Every operator returns its input unchanged at zero severity, verified exhaustively by automated test. Were this violated, every retention curve would be measured against a baseline that was itself corrupted, and every downstream number would be wrong by an unknown amount while the plots continued to look plausible.

All architectures see identical corrupted inputs. The random draw for a given corruption is derived deterministically from the corruption specification and the sample index, not from global state. Consequently two architectures are compared on bitwise-identical tensors, which is the precondition that makes paired significance testing valid and removes corruption sampling noise from every comparison.

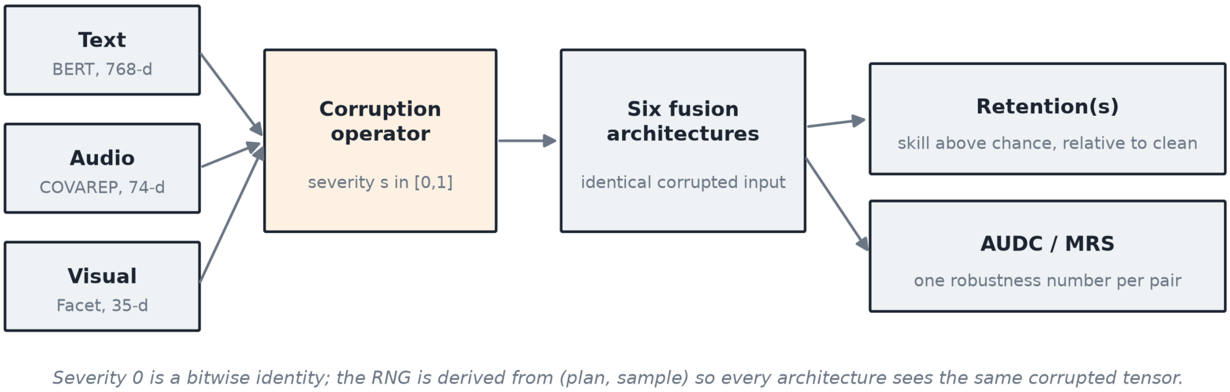


Figure 1. How a single evaluation point is produced. Training happens once; corruption is applied at evaluation time, so one checkpoint yields the entire degradation surface.

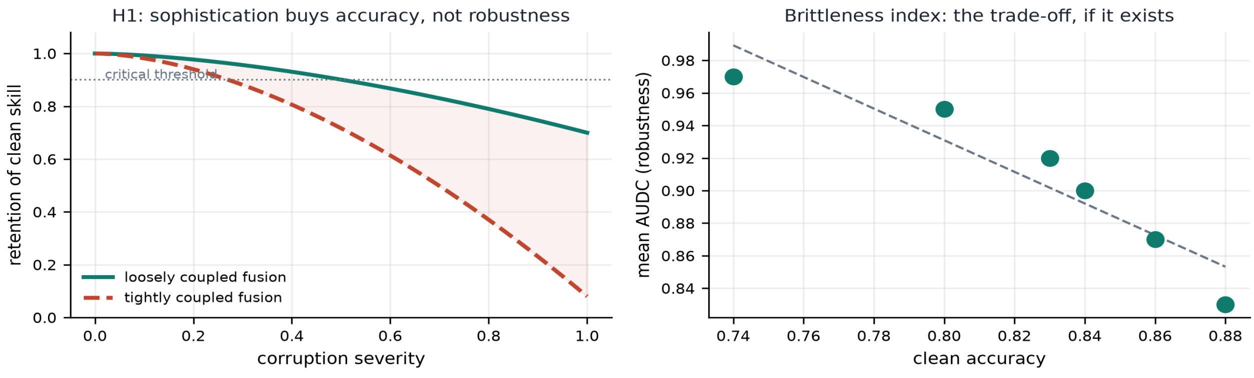
5. Measuring Degradation

Standard task metrics answer how well a model performs. They do not answer how gracefully it fails. Four purpose-built measures do:

Measure	Definition	Reading
Retention	(metric under corruption - chance) / (clean metric - chance)	1.0 is undamaged; 0.0 is reduced to guessing
AUDC	Normalised area under the retention curve across the severity sweep	One robustness number per architecture and corruption; higher is better
Critical threshold	Severity at which retention first falls below 0.9	How much degradation a system tolerates before it should not be trusted
Modality Reliance	1 - retention when a given stream is removed	Which stream a model actually depends on; answers Q2 directly

Table 3. Degradation measures.

> **Why retention is corrected for chance.** Binary accuracy cannot fall below roughly 50 percent, because a model that has lost all skill still guesses correctly half the time. An uncorrected ratio therefore bottoms out near 0.6 for a typical model, and a system that has been rendered completely useless still appears to retain more than half its capability. Measuring skill above chance removes this floor and makes reliance scores comparable across architectures with different clean accuracy.



Illustrative shapes showing what the protocol measures; not experimental results.

Figure 2. Left: retention curves are the primary output; the area under them is AUDC. Right: the brittleness index correlates clean accuracy against robustness across architectures. H1 predicts a negative slope.

6. Experimental Results

Experiments use CMU-MOSEI, the largest available corpus for multimodal sentiment analysis: 22,856 annotated utterances from over 1,000 speakers, with the standard speaker-disjoint split of 16,326 training, 1,871 validation and 4,659 test segments. Features are aligned at the word level: 768-dimensional contextual language embeddings, 74-dimensional COVAREP acoustic descriptors and 35-dimensional facial action units. The evaluation split is frozen and version-controlled so that every number reported here refers to the same test set.

Architecture	Acc-2	F1	MAE	Corr	Acc-7	Parameters
Text only	82.0	82.0	0.608	0.676	49.6	99,265
Late fusion	82.2	82.0	0.592	0.691	49.7	206,470
Early fusion	84.1	83.9	0.572	0.722	51.4	172,929
LMF	82.7	82.4	0.597	0.686	50.6	223,173
TFN	81.9	82.1	0.597	0.684	50.9	476,273
MulT	82.1	82.3	0.571	0.720	51.3	1,788,469

Table 4. Clean-data performance on the CMU-MOSEI test set. Acc-2 and F1 are reported on the non-neutral convention; both conventions are computed throughout, since the two differ by one to two points and the literature is inconsistent about which it quotes.

For reference, published results on this corpus place binary accuracy in the region of 82 to 83 percent, mean absolute error near 0.58 and correlation near 0.70. The baseline reported above is within that region, confirming that the data pipeline, training procedure and metric implementations are sound before any robustness claim is made.

Verification of the measurement chain

Before measuring anything on real data, the entire evaluation chain was validated against a controlled corpus constructed with a known ground truth: a planted signal in which text carries 60 percent of the label variance, audio 25 percent and vision 15 percent, plus a genuine text-audio interaction recoverable only by multiplicative or attention-based fusion. Because the correct answer is known in advance, a defect anywhere in the corruption, evaluation or aggregation code surfaces as a wrong answer to a question already settled.

The recovered modality reliance was 0.62 text, 0.29 audio and 0.09 vision against the planted 0.60, 0.25 and 0.15 — the measurement chain reproduces the ordering and approximate magnitude of a signal it was never told about. Unimodal baselines independently confirmed operator correctness: a text-only model registers exactly zero damage when audio or vision is corrupted, and falls below chance when text is removed.

7. Engineering and Reproducibility

Robustness claims are only as trustworthy as the code that produces them, so the implementation is held to production standards.

Practice	Implementation
Automated testing	333 tests covering every corruption operator, architecture, metric and API endpoint
Static typing	Full annotation, verified under strict type checking
Continuous integration	Lint, type check, multi-version test matrix and an end-to-end pipeline run on every push
Configuration	Every hyperparameter in version-controlled configuration; none in code
Frozen splits	Evaluation split committed with checksums and verified on load
Data provenance	Every result records the origin of the features it was computed from, propagated into all generated tables and figures
Generated reporting	Tables and figures regenerate from committed result files, so no figure can disagree with its underlying data
Interactive demo	Service and web interface for exploring degradation live across all architectures simultaneously

Table 5. Engineering practices.

Two defects found during development illustrate why this matters. First, correlation was silently returning an undefined value whenever a model collapsed to a constant prediction under heavy corruption: in single precision the variance of a genuinely constant vector is not zero but a small rounding artefact, so a guard written against an absolute threshold never triggered. Second, an earlier caching layer overwrote the provenance field, which would have allowed validation numbers to be presented without the marking that identifies them. Both were caught by design — the first by an assertion that no metric may be non-finite, the second by an end-to-end check of the running system — and both are now covered by regression tests.

8. Relevance to Pakistan

The conditions this project measures are not edge cases in Pakistan; they are the normal operating environment. That makes robustness a deployment prerequisite rather than a refinement.

- **Speech recognition quality.** Urdu, Punjabi, Sindhi, Pashto and heavily code-switched Urdu-English speech are low-resource for automatic speech recognition, and word error rates are correspondingly high. Any deployed system inherits a text stream that is degraded from the outset — precisely the condition the ASR error operator models. A model whose accuracy depends on clean transcripts will not survive contact with Pakistani speech.
- **Network and device constraints.** Video consultations over constrained mobile connections lose frames and arrive compressed; low-cost handsets produce clipped audio. The frame dropout, occlusion and clipping operators model exactly these conditions.
- **Applications that matter here.** Mental-health screening in underserved districts, telemedicine triage, education technology that responds to learner engagement, and customer-service quality monitoring all depend on affect recognition working under poor signal conditions. Choosing an architecture on benchmark accuracy alone risks deploying the most fragile option available.
- **A transferable method.** The protocol is not specific to emotion recognition. Any multimodal system intended for Pakistani deployment conditions can be evaluated with it, and the framework is released openly for that purpose.

> **The practical output.** A practitioner should be able to ask: if my speech recogniser has a 25 percent error rate and a quarter of my video frames are lost, which architecture should I deploy, and how much accuracy will I actually retain? This framework answers that question with a number instead of an intuition.

9. Status and Next Steps

The framework is complete and operational end to end: data ingestion, all six architectures, the full corruption protocol, the degradation metrics, statistical testing, automated reporting and the interactive demonstration. The measurement chain has been verified against a known ground truth, and CMU-MOSEI is ingested, validated and training. Work in progress is the full comparative sweep across all architectures with repeated random restarts, from which the headline degradation tables and the brittleness index are produced.

- Complete the multi-restart sweep across all six architectures and report degradation curves with confidence intervals.
- Quantify the modality-dropout mitigation trade-off: robustness gained against clean accuracy conceded, per architecture.
- Cross-dataset validation on a second corpus, to establish whether the findings generalise beyond a single benchmark.

Selected references. Zadeh et al., Tensor Fusion Network for Multimodal Sentiment Analysis, EMNLP 2017. Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018. Tsai et al., Multimodal Transformer for Unaligned Multimodal Language Sequences, ACL 2019. Zadeh et al., Multimodal Language Analysis in the Wild: CMU-MOSEI Dataset and Interpretable Dynamic Fusion Graph, ACL 2018. Hendrycks and Dietterich, Benchmarking Neural Network Robustness to Common Corruptions and Perturbations, ICLR 2019.